Supplementary Material for Towards Reliable Multi-Label Classification via Conditional Dependency Modeling

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1 Proof of Theorem 1

Proof. We shall begin with the relation between Kullback-Leibler divergence and Information matrix.
Let f_α be any probability distribution with parameter α and slightly perturbed distribution is $f_{\alpha+\delta}$. Then it is known that if the KL divergence admits a quadratic expansion then

$$D_{KL}(f_\alpha || f_{\alpha+\delta}) \approx \frac{1}{2}\delta^\top \mathcal{I}(\alpha)\delta^\top \tag{1}$$

From Larralde et al. [1] we can say that the joint distribution can be expressed as the perturbation around the product of marginals if the moments exist. Hence, we can say

$$\prod_i P(Z^{(i)};\alpha) = P(\mathbf{Z};\alpha+\delta) \text{ where } Z^{(i)} = Y^{(i)}|\mathbf{X} \text{ and } \mathbf{Z} = \mathbf{Y}|\mathbf{X}$$

Now, by using equation 1 we have

$$D_{KL}(P(\mathbf{Z};\alpha)||P(\mathbf{Z};\alpha+\delta)) \approx \frac{1}{2}\delta^\top \mathcal{I}(\alpha)\delta^\top$$

Since we are dealing with multi-label dataset $\mathbf{Z} \in \{0,1\}^L$, means the exact distribution is complex (2^L states). So we shall use Ising Model for our approximation. This is the standard mathematical tool for modelling binary variables with pairwise correlations. It can be written as

$$P(\mathbf{Z};\alpha) \approx \frac{1}{G(\alpha)} \exp\left(\sum_i \alpha_i z^{(i)} + \sum_{i<j} \alpha_{ij} z^{(i)} z^{(j)}\right) \tag{2}$$

α_i are the parameters corresponding to label i and α_{ij} is the interaction parameter between label i and j. $G(\alpha)$ is the partition function. Now we take the log-likelihood

$$\mathcal{F} = \log P(\mathbf{Z};\alpha) = \sum_i \alpha_i z^{(i)} + \sum_{i<j} \alpha_{ij} z^{(i)} z^{(j)} - \log G(\alpha)$$

Differentiate with respect to a specific parameter α_m:

$$\frac{\partial \mathcal{F}}{\partial \alpha_m} = z^{(m)} - \frac{\partial \log G(\alpha)}{\partial \alpha_u}$$

From exponential family theory, the derivative of the log-partition function is the expected value:

$$\frac{\partial \mathcal{F}}{\partial \alpha_m} = z^{(m)} - \mathbb{E}[z^{(m)}]$$

Now, differentiate again with respect to a different parameter α_n. Since $z^{(m)}$ is data (constant), its derivative is zero. We only differentiate the expectation term:

$$\frac{\partial^2 \mathcal{F}}{\partial \alpha_n \partial \alpha_m} = -\frac{\partial \mathbb{E}[z^{(m)}]}{\partial \alpha_n}$$

For exponential family the derivative of the mean of one variable with respect to the canonical parameter of another is exactly their Covariance. Hence,

$$\frac{\partial \mathbb{E}[z^{(m)}]}{\partial \alpha_n} = \mathrm{Cov}(z^{(m)}, z^{(n)})$$

By substituting this into the Fisher information we get

$$\begin{aligned} \mathcal{I}(\alpha)_{mn} &= -\mathbb{E}\left[\frac{\partial^2 \mathcal{F}}{\partial \alpha_m \partial \alpha_n}\right] = -\mathbb{E}\left[-\mathrm{Cov}(z^{(m)}, z^{(n)})\right] \\ &= \mathrm{Cov}(z^{(m)}, z^{(n)}) \end{aligned}$$

So we have established that $D_{KL}(P(\mathbf{Z};\alpha)||P(\mathbf{Z};\alpha+\delta)) \approx \mathrm{Cov}(z^{(m)}, z^{(n)}), \forall\, m, n \in \{1, \ldots, L\}$. The conditional dependency is always proportional to the covariance between the occurrence of labels hence, $\mathrm{Cov}(z^{(m)}, z^{(n)}) \propto \mathrm{Cov}(y^{(m)}, y^{(n)})$.

2 Labels and Correlations

We visualize the correlation structure among the five most frequent classes in the MS-COCO, PASCAL VOC 2012, and WIDER-A datasets. The predictions are obtained using the ViT-B/32 architecture. From Figure 1 it can be observed that the correlations induced by the proposed CMLL loss closely match the correlation patterns present in the ground-truth labels. This observation suggests that CMLL effectively captures the underlying conditional dependencies among labels during training.



(a) MS-COCO GT **(b)** MS-COCO Predictions

(c) PASCAL VOC 2012 GT **(d)** PASCAL VOC 2012 Predictions

(e) WIDER-A GT **(f)** WIDER-A Predictions

Fig. 1: Correlation among the five most frequent classes in the MS-COCO, PASCAL VOC 2012, and WIDER-A dataset. Pos → Positively Correlated; Neg → Negatively Correlated; GT → Ground Truth.

References

1. Larralde, H.: Maximum-entropy distributions of correlated variables with prespecified marginals. Physical Review E—Statistical, Nonlinear, and Soft Matter Physics **86**(6), 061117 (2012)